



Lecture 2: Mathematical Models

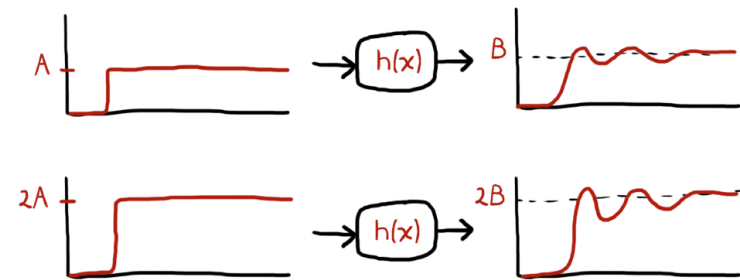
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SIST 1D#206



Linear System

- Linearity is defined in terms of the input $u(t)$ and output $y(t)$
 - Also known as excitation and response.
- If a system satisfies **homogeneity** and **superposition**, it is a linear system.
- If a linear system further sanctifies the property of **time-invariance**, it is then called an LTI system.
 - Linearity + Time-invariance = **sinusoidal fidelity**
 - Linear system is a wider concept than *linear time-invariant (LTI)* system, also including *linear time-varying* system

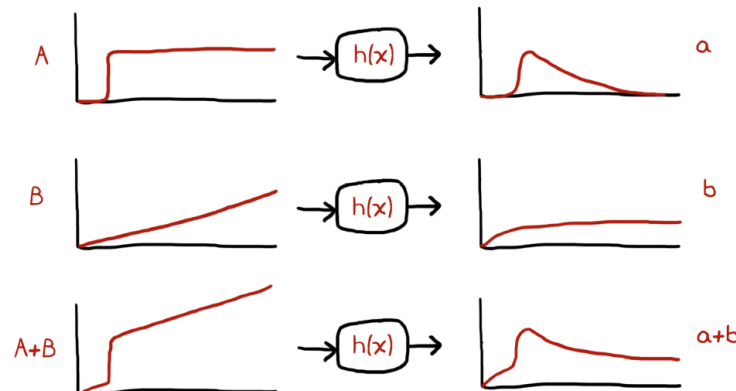
Homogeneity



Homogeneity

$$h(ax) = a \cdot h(x)$$

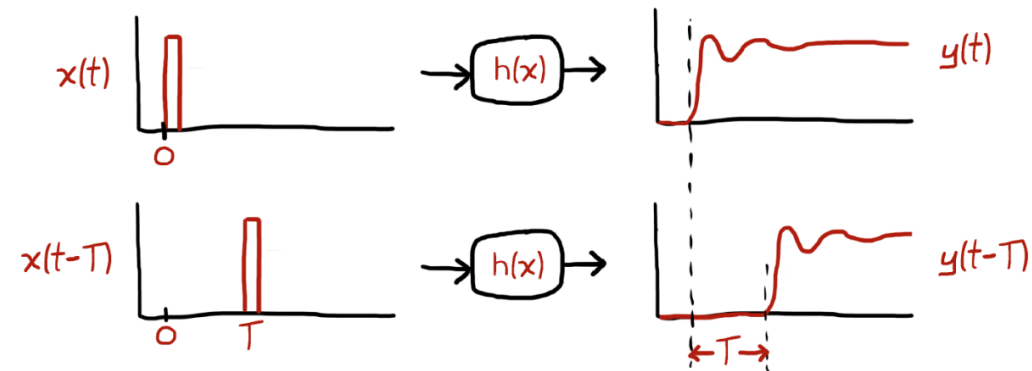
Superposition



Superposition

$$h(x_1) + h(x_2) = h(x_1 + x_2)$$

Time Invariance



- Linear time-invariant operator

LTI Allowable operations:

Multiply or divide the input by a constant

$$a \cdot x(t)$$

$$\frac{1}{a} \cdot x(t)$$

Integrate or differentiate the input

$$\int x(t) dt$$

$$\frac{dx(t)}{dt}$$

Add or subtract multiple inputs

$$x_1(t) + x_2(t)$$

$$x_1(t) - x_2(t)$$

- Generally speaking, a system is also an operator (we are not ready to see this yet)

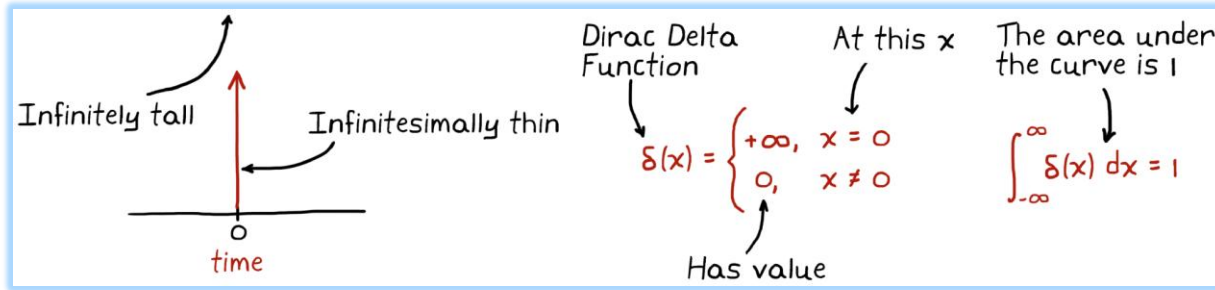
True or False:

The differential equation $\frac{d}{dt}y = -ty + u$ describes a linear system.



Laplace Transform

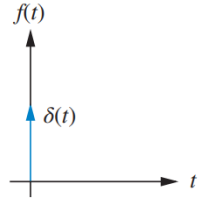
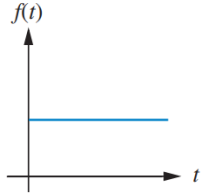
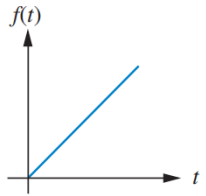
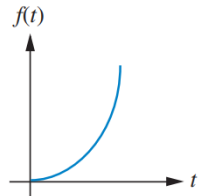
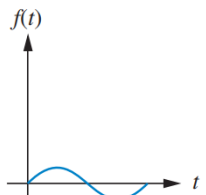
Dirac delta function $\delta(t)$



Heaviside step function $u(t)$

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

TABLE 1.1 Test waveforms used in control systems

Input	Function	Description	Sketch	Use
Impulse	$\delta(t)$	$\delta(t) = \infty$ for $0- < t < 0+$ $= 0$ elsewhere $\int_{0-}^{0+} \delta(t) dt = 1$		Transient response Modeling
Step	$u(t)$	$u(t) = 1$ for $t > 0$ $= 0$ for $t < 0$		Transient response Steady-state error
Ramp	$tu(t)$	$tu(t) = t$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Parabola	$\frac{1}{2}t^2u(t)$	$\frac{1}{2}t^2u(t) = \frac{1}{2}t^2$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Sinusoid	$\sin \omega t$			Transient response Modeling Steady-state error

- Laplace transform is an integral transform with a fast decaying kernel function,
 - because the most important thing for an integral is to exist

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$

Kernel is: $K(s, t) = e^{-st}$

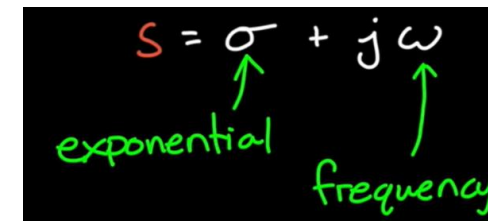


TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$ $u(t) = 1 \quad t > 0$ $= 0 \quad t < 0$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Example 2.1

Laplace Transform of a Time Function

PROBLEM: Find the Laplace transform of $f(t) = Ae^{-at}u(t)$.

SOLUTION: Since the time function does not contain an impulse function, we can replace the lower limit of Eq. (2.1) with 0. Hence,

$$\begin{aligned}
 F(s) &= \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} Ae^{-at}e^{-st} dt = A \int_0^{\infty} e^{-(s+a)t} dt \\
 &= -\frac{A}{s+a} e^{-(s+a)t} \Big|_{t=0}^{\infty} = \frac{A}{s+a} \quad (2.3)
 \end{aligned}$$

❑ Inverse Laplace transform is often done with further help of the following table

Example 2.2

Inverse Laplace Transform

PROBLEM: Find the inverse Laplace transform of $F_1(s) = 1/(s + 3)^2$.

SOLUTION: For this example we make use of the frequency shift theorem, Item 4 of Table 2.2, and the Laplace transform of $f(t) = tu(t)$, Item 3 of Table 2.1. If the inverse transform of $F(s) = 1/s^2$ is $tu(t)$, the inverse transform of $F(s + a) = 1/(s + a)^2$ is $e^{-at}tu(t)$. Hence, $f_1(t) = e^{-3t}tu(t)$.

TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t - T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{k-1}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

□ The two tables can be used to derive for more complicated results, e.g.,

$$F_1(s) = \frac{s^3 + 2s^2 + 6s + 7}{s^2 + s + 5}$$

$$F_1(s) = s + 1 + \frac{2}{s^2 + s + 5}$$

$$f_1(t) = \frac{d\delta(t)}{dt} + \delta(t) + \mathcal{L}^{-1}\left[\frac{2}{s^2 + s + 5}\right]$$

TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem

$$\mathcal{L}[Ae^{-at}\cos \omega t] = \frac{A(s + a)}{(s + a)^2 + \omega^2}$$

- Laplace transform converts a differential equation into algebraic equation for its **differentiation theorem**

TABLE 2.2 Laplace transform theorems

Item no.	Theorem		Name
7.	$\mathcal{L} \left[\frac{df}{dt} \right]$	$= sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L} \left[\frac{d^2f}{dt^2} \right]$	$= s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L} \left[\frac{d^nf}{dt^n} \right]$	$= s^nF(s) - \sum_{k=1}^n s^{n-k}f^{k-1}(0-)$	Differentiation theorem

Example 2.3

Laplace Transform Solution of a Differential Equation

PROBLEM: Given the following differential equation, solve for $y(t)$ if all initial conditions are zero. Use the Laplace transform.

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 32y = 32u(t) \quad (2.14)$$

Laplace Transform



PROBLEM: Find the Laplace transform of $f(t) = te^{-5t}$.

ANSWER:



TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$ $u(t) = 1 \quad t > 0$ $= 0 \quad t < 0$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

PROBLEM: Find the inverse Laplace transform of $F(s) = 10/[s(s+2)(s+3)^2]$.

ANSWER:



TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t-T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{(k-1)}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²



Transfer Function

- General case of a high order differential equation:

$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t) \quad (2.50)$$

- **Convert it into s-domain:**

$$\begin{aligned} a_n s^n C(s) + a_{n-1} s^{n-1} C(s) + \cdots + a_0 C(s) + \text{initial condition} \\ \text{terms involving } c(t) \\ = b_m s^m R(s) + b_{m-1} s^{m-1} R(s) + \cdots + b_0 R(s) + \text{initial condition} \\ \text{terms involving } r(t) \end{aligned}$$

- Assume zero initial conditions we get transfer function

$$\frac{C(s)}{R(s)} = G(s) = \frac{(b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0)}{(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0)}$$

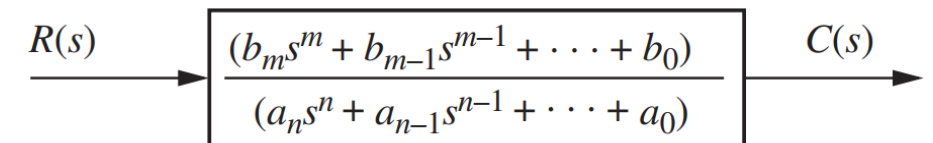


FIGURE 2.2 Block diagram of a transfer function

PROBLEM: Find the transfer function, $G(s) = C(s)/R(s)$, corresponding to the differential equation $\frac{d^3 c}{dt^3} + 3\frac{d^2 c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2 r}{dt^2} + 4\frac{dr}{dt} + 3r$.

ANSWER:

PROBLEM: Find the differential equation corresponding to the transfer function,

$$G(s) = \frac{2s + 1}{s^2 + 6s + 2}$$

ANSWER:

PROBLEM: Find the ramp response for a system whose transfer function is

$$G(s) = \frac{s}{(s + 4)(s + 8)}$$

ANSWER:



System Examples

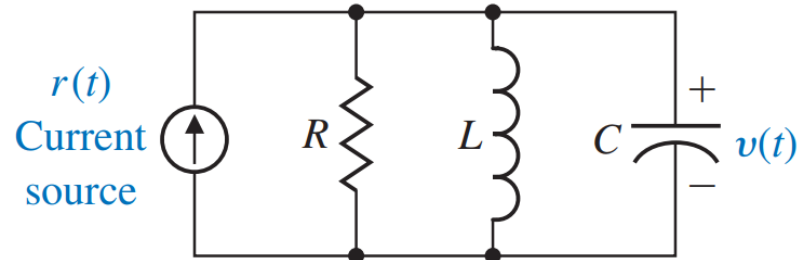
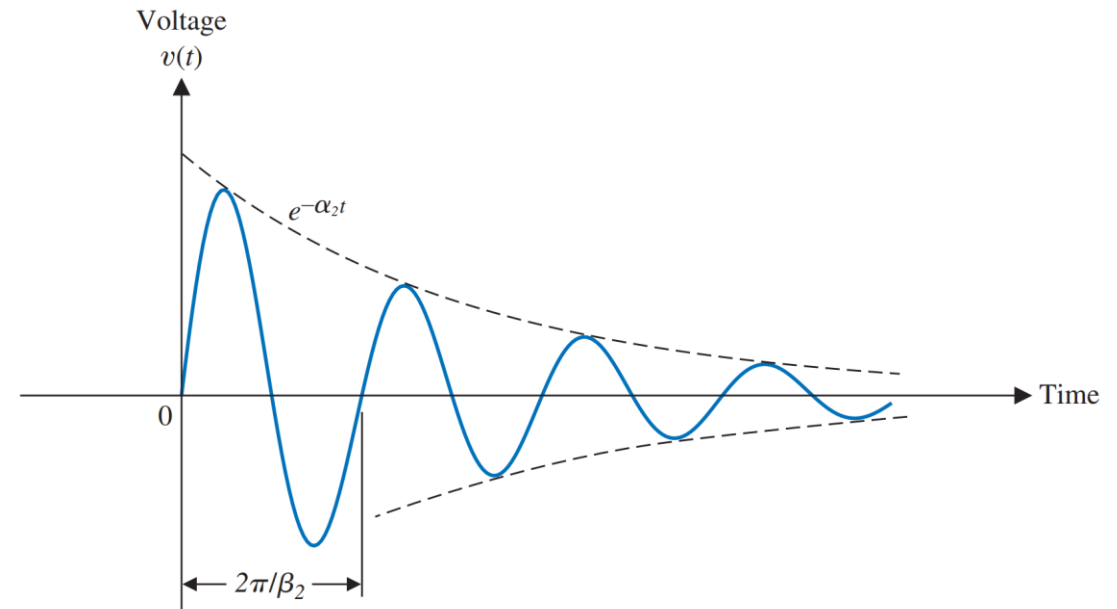


FIGURE 2.3
RLC circuit.

$$\frac{v(t)}{R} + C \frac{dv(t)}{dt} + \frac{1}{L} \int_0^t v(t) dt = r(t)$$

$$v(t) = K_2 e^{-\alpha_2 t} \cos(\beta_2 t + \theta_2)$$

FIGURE 2.4
Typical voltage
response for
an *RLC* circuit.



Spring mass damper

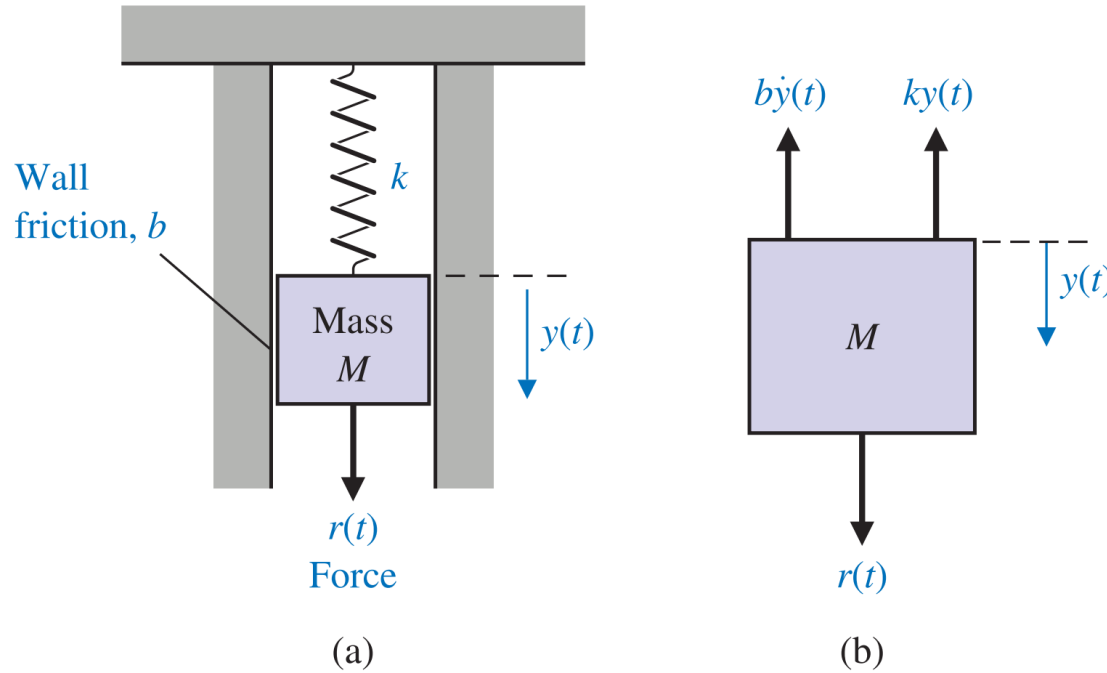


FIGURE 2.2
(a) Spring-mass-damper system.
(b) Free-body diagram.

$$M \frac{d^2 y(t)}{dt^2} + b \frac{dy(t)}{dt} + ky(t) = r(t)$$

$$y(t) = K_1 e^{-\alpha_1 t} \sin(\beta_1 t + \theta_1)$$

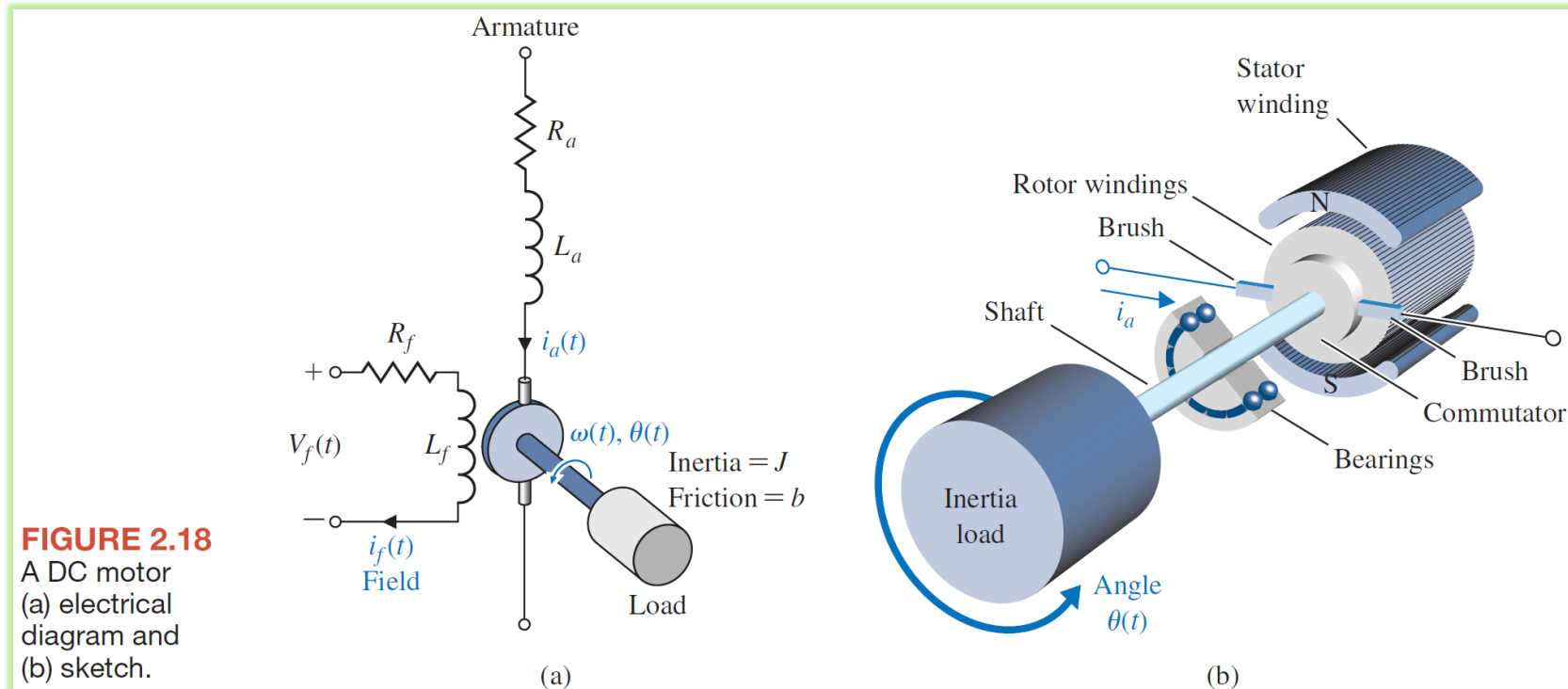
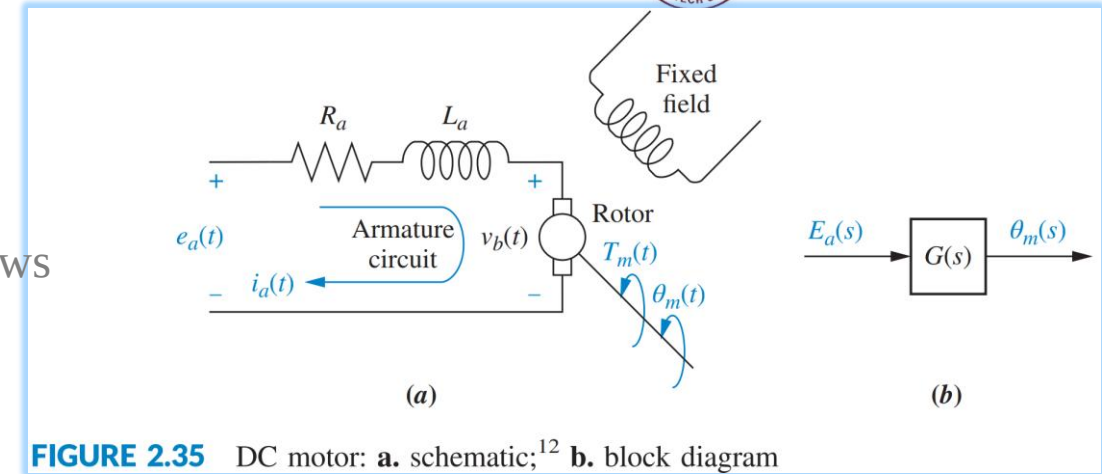
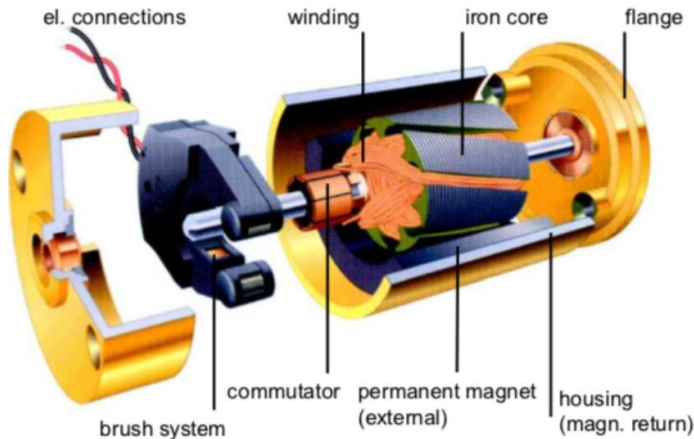
Electro-mechanical system

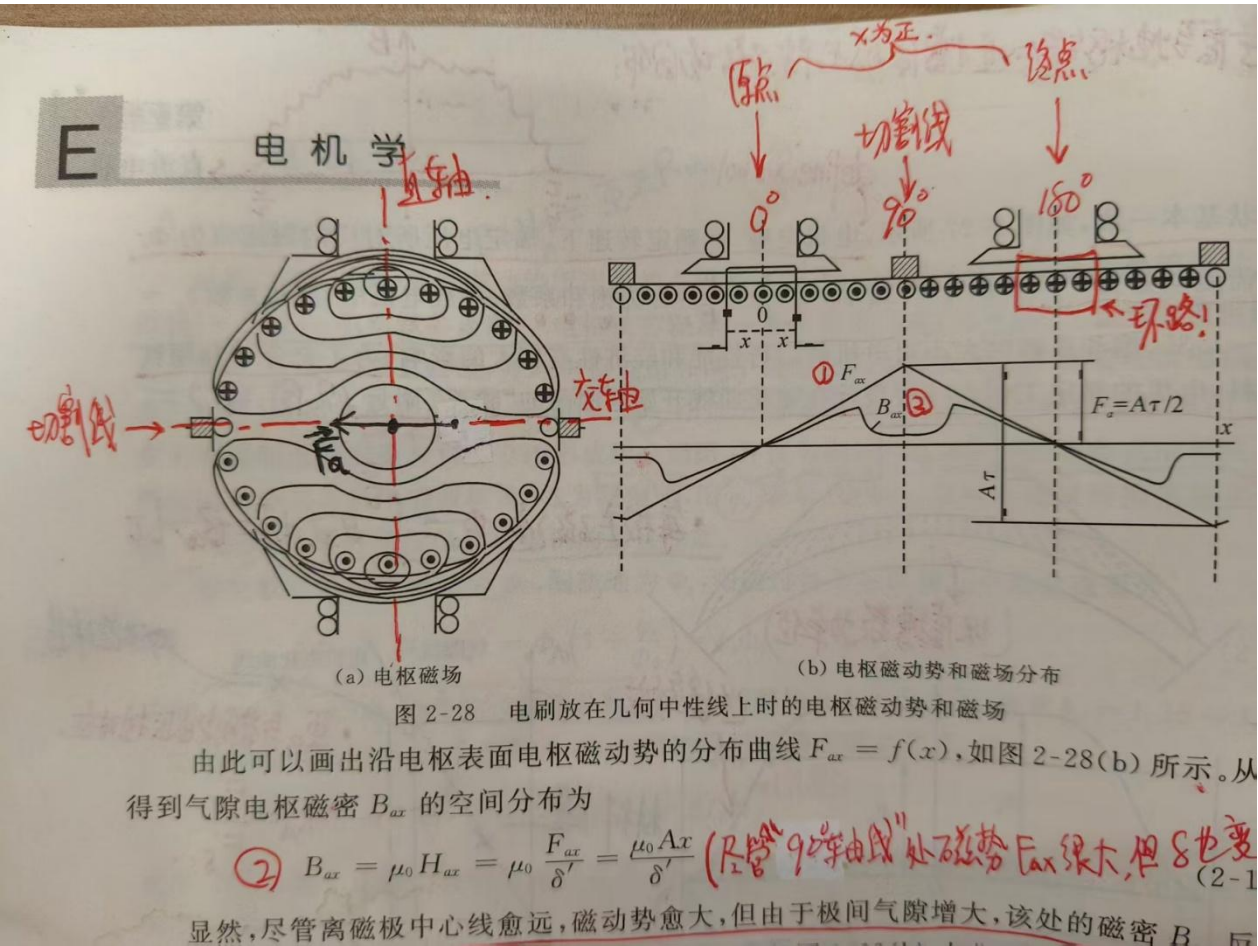


- ❑ Newtonian mechanics
- ❑ Faraday's law of induction
 - With a little bit of math of total differential and derivative laws
- ❑ Ampere force

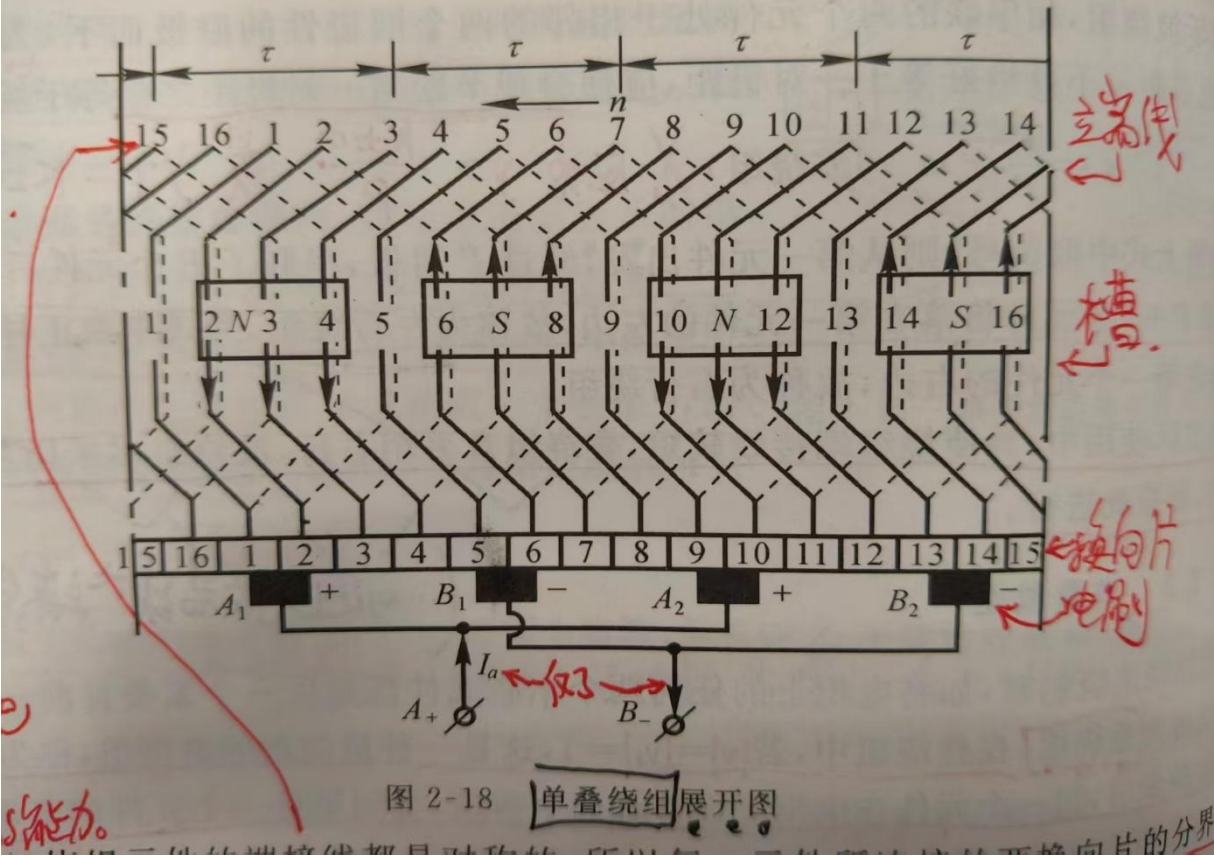
$$J \frac{d^2}{dt^2} \Theta = D \frac{d}{dt} \Theta + K_T i_1$$

$$u_1 = R i_1 + K_E \frac{d}{dt} \Theta + L_1 \frac{d}{dt} i_1$$



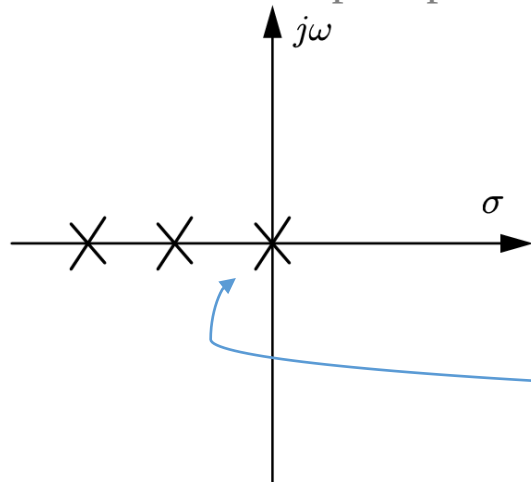


应电势方向和电刷电位的正负极性如图 2-18 所示。



Overview of the Math Models

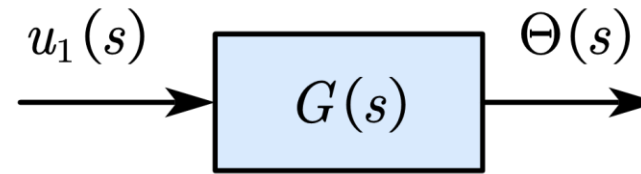
- ❑ O.D.E (from physics laws)
- ❑ Block diagram
- ❑ Transfer function (via Laplace transform)
- ❑ State space model
- ❑ Pole zero map (related to eigenvalues)
 - Root locus and pole placement



Eigenvalues
(when input is set to 0)

$$J \frac{d^2}{dt^2} \Theta = D \frac{d}{dt} \Theta + K_T i_1$$

$$u_1 = R i_1 + K_E \frac{d}{dt} \Theta + L_1 \frac{d}{dt} i_1$$



Can you give details within $G(s)$?

$$\frac{d}{dt} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & K_T J^{-1} \\ 0 & -K_E L^{-1} & -R L^{-1} \end{bmatrix} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ L^{-1} \end{bmatrix} u + \begin{bmatrix} 0 \\ J^{-1} \\ 0 \end{bmatrix} \tau_L$$



Exercise for modeling of an e.m. system



Example 2.23

Transfer Function—DC Motor and Load

PROBLEM: Given the system and torque-speed curve of Figure 2.39(a) and (b), find the transfer function, $\theta_L(s)/E_a(s)$.

SOLUTION: Begin by finding the mechanical constants, J_m and D_m , in Eq. (2.153). From Eq. (2.155), the total inertia at the armature of the motor is

$$J_m = J_a + J_L \left(\frac{N_1}{N_2} \right)^2 = 5 + 700 \left(\frac{1}{10} \right)^2 = 12 \quad (2.164)$$

and the total damping at the armature of the motor is

$$D_m = D_a + D_L \left(\frac{N_1}{N_2} \right)^2 = 2 + 800 \left(\frac{1}{10} \right)^2 = 10 \quad (2.165)$$

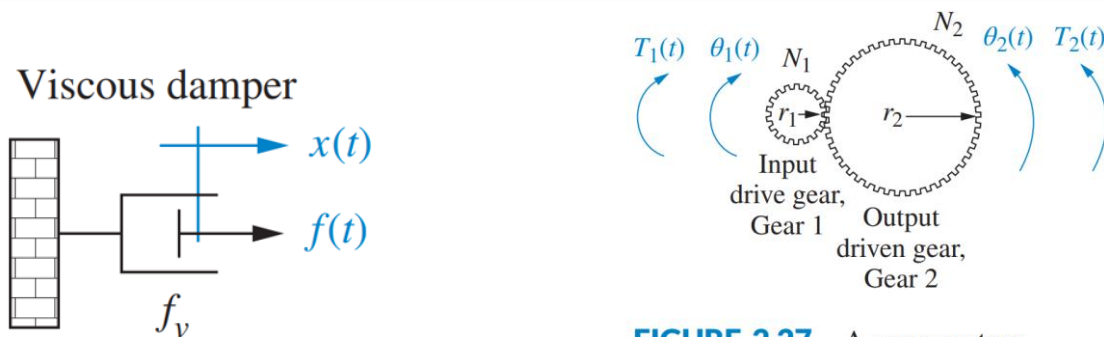


FIGURE 2.27 A gear system

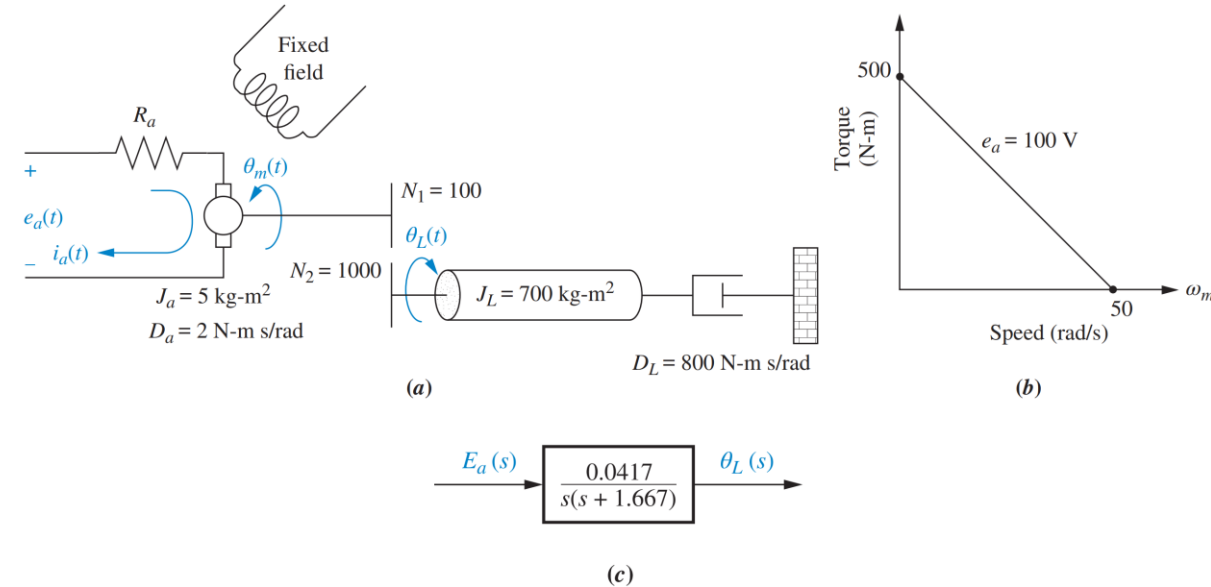


FIGURE 2.39 a. DC motor and load; b. torque-speed curve; c. block diagram

Generalizing the results, we can make the following statement: *Rotational mechanical impedances can be reflected through gear trains by multiplying the mechanical impedance by the ratio*

$$\left(\frac{\text{Number of teeth of gear on destination shaft}}{\text{Number of teeth of gear on source shaft}} \right)^2$$

where the impedance to be reflected is attached to the source shaft and is being reflected to the destination shaft. The next example demonstrates the application of the concept of reflected impedances as we find the transfer function of a rotational mechanical system with gears.

- ❑ The simultaneous **first-order differential** equations about the state variables
 - State variables must be linearly independent; that is, no state variable can be written as a linear combination of the other state variables.

- ❑ **Algebraic** output equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

$\mathbf{x}$ = state vector

$\dot{\mathbf{x}}$ = derivative of the state vector with respect to time

$\mathbf{y}$ = output vector

$\mathbf{u}$ = input or control vector

$\mathbf{A}$ = system matrix

$\mathbf{B}$ = input matrix

$\mathbf{C}$ = output matrix

$\mathbf{D}$ = feedforward matrix

- ❑ E.g.,:

$$\frac{d}{dt} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & K_T J^{-1} \\ 0 & -K_E L^{-1} & -R L^{-1} \end{bmatrix} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ L^{-1} \end{bmatrix} u + \begin{bmatrix} 0 \\ J^{-1} \\ 0 \end{bmatrix} \tau_L$$

$$y = [1 \ 0 \ 1] \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix}$$

- ❑ State space: The n-dimensional space whose axes are the state variables.

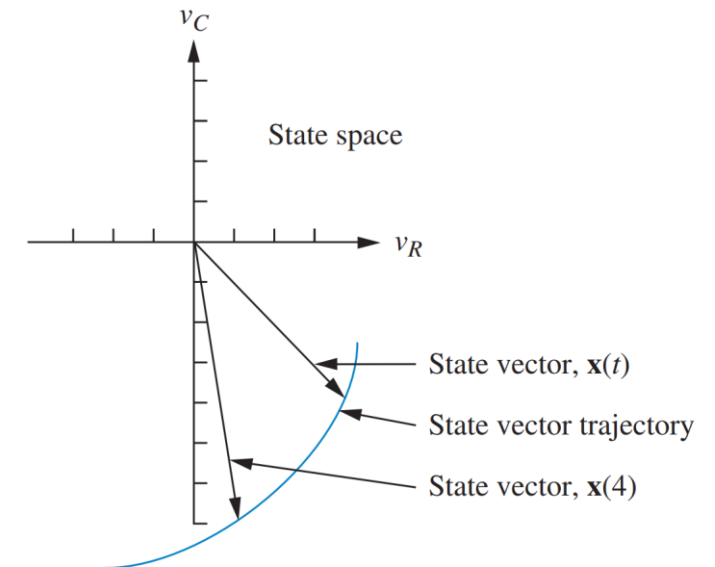


FIGURE 3.3 Graphic representation of state space and a state vector

PROBLEM: Find the state-space representation of the electrical network shown in Figure 3.8. The output is $v_o(t)$.

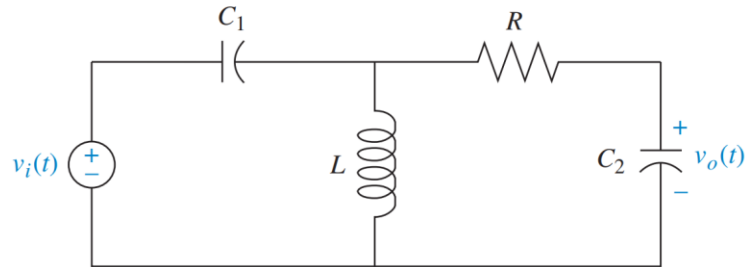


FIGURE 3.8 Electric circuit for Skill-Assessment Exercise 3.1

ANSWER:



PROBLEM: Represent the translational mechanical system shown in Figure 3.9 in state space, where $x_3(t)$ is the output.

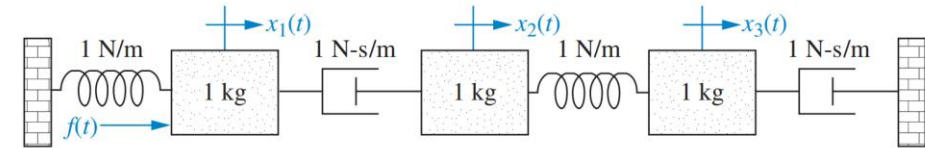


FIGURE 3.9 Translational mechanical system for Skill-Assessment Exercise 3.2

ANSWER:



From O.D.E to State space model



At first we select a set of state variables, called *phase variables*, where each subsequent state variable is defined to be the derivative of the previous state variable.

$$\frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 u$$



$$\begin{aligned} x_1 &= y \\ x_2 &= \frac{dy}{dt} \\ x_3 &= \frac{d^2 y}{dt^2} \\ &\vdots \\ x_n &= \frac{d^{n-1} y}{dt^{n-1}} \end{aligned}$$



$$\begin{aligned} \dot{x}_1 &= \frac{dy}{dt} \\ \dot{x}_2 &= \frac{d^2 y}{dt^2} \\ \dot{x}_3 &= \frac{d^3 y}{dt^3} \\ &\vdots \\ \dot{x}_n &= \frac{d^n y}{dt^n} \end{aligned}$$



$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \end{aligned}$$



$$\dot{x}_n = -a_0 x_1 - a_1 x_2 \cdots - a_{n-1} x_n + b_0 u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 & -a_4 & -a_5 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} u \quad (3.52)$$

$$y = [1 \quad 0 \quad 0 \quad \cdots \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

Converting a Transfer Function with a Constant Term in the Numerator

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$(s^3 + 9s^2 + 26s + 24)C(s) = 24R(s)$$

$$\ddot{c} + 9\dot{c} + 26c = 24r$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

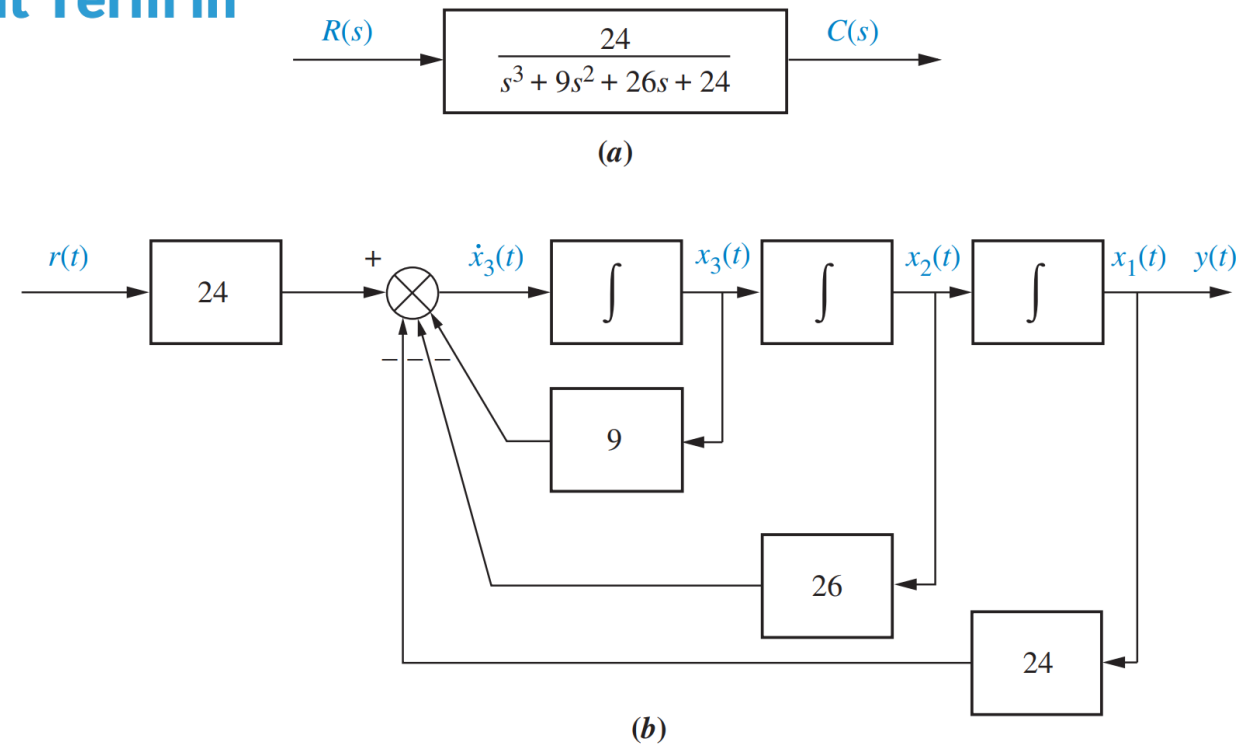


FIGURE 3.10 a. Transfer function; b. equivalent block diagram showing phase variables. Note: $y(t) = c(t)$.

Converting a Transfer Function with a Polynomial in the Numerator

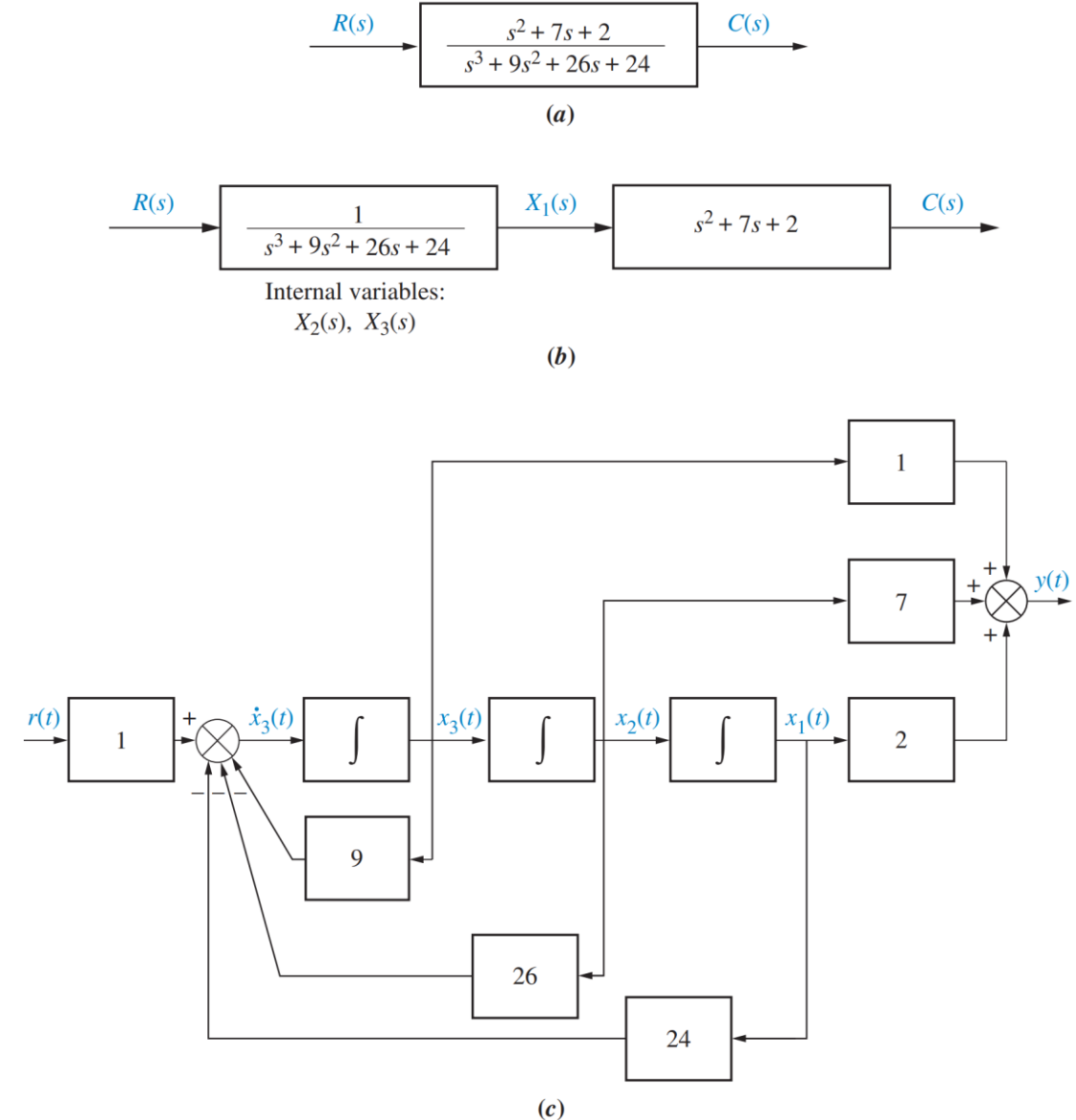


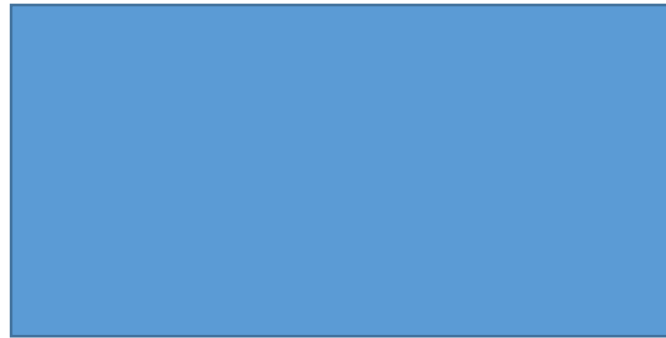
FIGURE 3.12 **a.** Transfer function; **b.** decomposed transfer function; **c.** equivalent block diagram Note: $y(t) = c(t)$.

From transfer function to state space model (2)



PROBLEM: Find the state equations and output equation for the phase-variable representation of the transfer function $G(s) = \frac{2s + 1}{s^2 + 7s + 9}$.

ANSWER:



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$



$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$



$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}\mathbf{U}(s)$$



$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)$$



$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) + \mathbf{D}\mathbf{U}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s)$$

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$



Single input single output

Transfer function matrix

State space model to transfer function (ss2tf)



PROBLEM: Given the system defined by Eq. (3.74), find the transfer function, $T(s) = Y(s)/U(s)$, where $U(s)$ is the input and $Y(s)$ is the output.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad (3.74a)$$

$$y = [1 \ 0 \ 0] \mathbf{x} \quad (3.74b)$$

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} = \frac{\begin{bmatrix} (s^2 + 3s + 2) & s + 3 & 1 \\ -1 & s(s + 3) & s \\ -s & -(2s + 1) & s^2 \end{bmatrix}}{s^3 + 3s^2 + 2s + 1} \quad (s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s + 3 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \text{adj}(\mathbf{A}) = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$
$$T(s) = \frac{10(s^2 + 3s + 2)}{s^3 + 3s^2 + 2s + 1}$$

TryIt 3.2

Use the following MATLAB and the Control System Toolbox statements to obtain the transfer function shown in Skill-Assessment Exercise 3.4 from the state-space representation of Eq. (3.78).

```
A=[ -4  -1.5; 4  0];  
B=[ 2  0]';  
C=[ 1.5  0.625];  
D=0;  
T=ss(A, B, C, D);  
T=tf(T)
```

PROBLEM: Convert the state and output equations shown in Eq. (3.78) to a transfer function.

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t) \quad (3.78a)$$

$$y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} \mathbf{x} \quad (3.78b)$$

ANSWER:



$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{adj}(\mathbf{A}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

State space model is not unique



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + \mathbf{D}u$$

$$\mathbf{z} = \mathbf{P}^{-1}\mathbf{x}$$



$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}u$$

$$y = \mathbf{C}\mathbf{P}\mathbf{z} + \mathbf{D}u$$

PROBLEM: Given the system represented in state space by Eqs. (5.73),

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad (5.73a)$$

$$y = [1 \ 0 \ 0] \mathbf{x} \quad (5.73b)$$

transform the system to a new set of state variables, $\mathbf{z}$, where the new state variables are related to the original state variables, $\mathbf{x}$, as follows:

$$\mathbf{z} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \mathbf{x} = \mathbf{P}^{-1}\mathbf{x} \quad (5.74a)$$

$$z_1 = 2x_1 \quad (5.74b)$$

$$z_2 = 3x_1 + 2x_2 \quad (5.74c)$$

$$z_3 = x_1 + 4x_2 + 5x_3 \quad (5.74c)$$

$$\begin{aligned} \mathbf{P}^{-1}\mathbf{A}\mathbf{P} &= \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix} \\ &= \begin{bmatrix} -1.5 & 1 & 0 \\ -1.25 & 0.7 & 0.4 \\ -2.5 & 0.4 & -6.2 \end{bmatrix} \end{aligned}$$

Diagonal state space representation

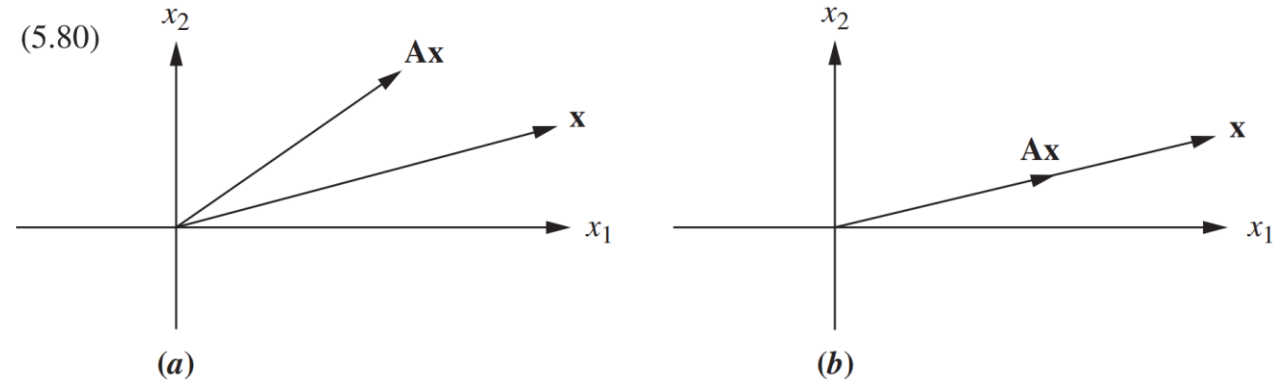


Eigenvector. The eigenvectors of the matrix $\mathbf{A}$ are all vectors, $\mathbf{x}_i \neq \mathbf{0}$, which under the transformation $\mathbf{A}$ become multiples of themselves; that is,

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i$$

where λ_i 's are constants.

FIGURE 5.32 To be an eigenvector, the transformation $\mathbf{A}\mathbf{x}$ must be collinear with $\mathbf{x}$; thus, in (a), $\mathbf{x}$ is not an eigenvector; in (b), it is.



Eigenvalue. The eigenvalues of the matrix $\mathbf{A}$ are the values of λ_i that satisfy Eq. (5.80) for $\mathbf{x}_i \neq \mathbf{0}$.

To find the eigenvectors, we rearrange Eq. (5.80). Eigenvectors, $\mathbf{x}_i$, satisfy

$$\mathbf{0} = (\lambda_i\mathbf{I} - \mathbf{A})\mathbf{x}_i \quad (5.81)$$

Solving for $\mathbf{x}_i$ by premultiplying both sides by $(\lambda_i\mathbf{I} - \mathbf{A})^{-1}$ yields

$$\mathbf{x}_i = (\lambda_i\mathbf{I} - \mathbf{A})^{-1}\mathbf{0} = \frac{\text{adj}(\lambda_i\mathbf{I} - \mathbf{A})}{\det(\lambda_i\mathbf{I} - \mathbf{A})}\mathbf{0} \quad (5.82)$$

Since $\mathbf{x}_i \neq \mathbf{0}$, a nonzero solution exists if

$$\det(\lambda_i\mathbf{I} - \mathbf{A}) = 0 \quad (5.83)$$

□ Diagonal representation

- Equivalent to partial fraction expansion of t.f.

$$\mathbf{P} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n]$$

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i$$

$$\mathbf{AP} = \mathbf{PD}$$

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$$

$$\mathbf{z} = \mathbf{P}^{-1}\mathbf{AP}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{P}\mathbf{z} + \mathbf{D}\mathbf{u}$$

from which λ_i , the eigenvalues, can be found.

Diagonal state space representation



Diagonalizing a System in State Space

PROBLEM: Given the system of Eqs. (5.94), find the diagonal system that is similar.

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u \quad (5.94a)$$

$$y = [2 \quad 3] \mathbf{x} \quad (5.94b)$$

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$\mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$$

$$\mathbf{C}\mathbf{P} = [2 \quad 3] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = [5 \quad -1]$$

$$\dot{\mathbf{z}} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix} u \quad \boxed{\mathbf{z} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}u}$$

$$y = [5 \quad -1] \mathbf{z} \quad \boxed{y = \mathbf{C}\mathbf{P}\mathbf{z} + \mathbf{D}u}$$

$$\begin{aligned} \det(\lambda\mathbf{I} - \mathbf{A}) &= \left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \right| \\ &= \begin{vmatrix} \lambda + 3 & -1 \\ -1 & \lambda + 3 \end{vmatrix} \\ &= \lambda^2 + 6\lambda + 8 \end{aligned}$$

$$\begin{aligned} \mathbf{A}\mathbf{x}_i &= \lambda\mathbf{x}_i \\ \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= -2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= -4 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} c \\ c \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} c \\ -c \end{bmatrix}$$

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Diagonal state space representation



PROBLEM: For the system represented in state space as follows:

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 3 \\ -4 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 4 \end{bmatrix} \mathbf{x}$$

convert the system to one where the new state vector, $\mathbf{z}$, is

$$\mathbf{z} = \begin{bmatrix} 3 & -2 \\ 1 & -4 \end{bmatrix} \mathbf{x}$$

ANSWER:

$$\dot{\mathbf{z}} = \begin{bmatrix} 6.5 & -8.5 \\ 9.5 & -11.5 \end{bmatrix} \mathbf{z} + \begin{bmatrix} -3 \\ -11 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0.8 & -1.4 \end{bmatrix} \mathbf{z}$$

PROBLEM: For the original system find the diagonal system that is similar.

ANSWER:

$$\dot{\mathbf{z}} = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 18.39 \\ 20 \end{bmatrix} u$$
$$y = \begin{bmatrix} -2.121 & 2.6 \end{bmatrix} \mathbf{z}$$

Basic components:

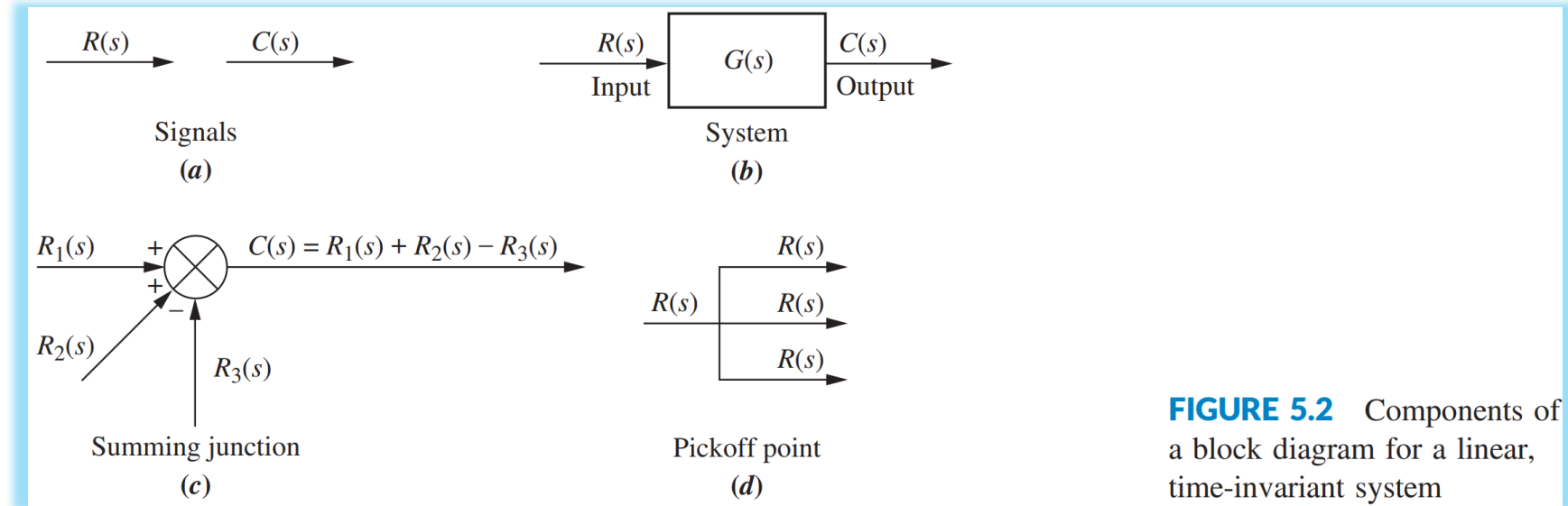


FIGURE 5.2 Components of a block diagram for a linear, time-invariant system

Cascade and parallel:

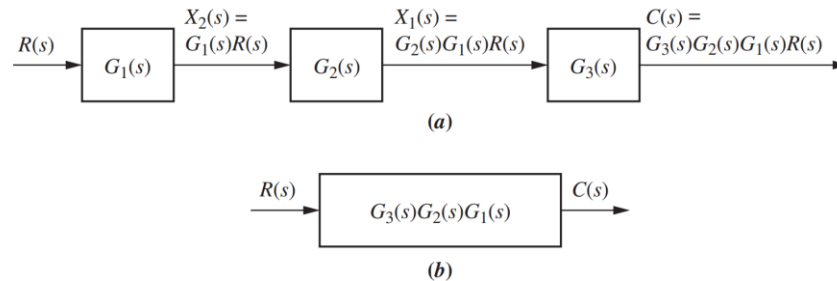


FIGURE 5.3 a. Cascaded subsystems; b. equivalent transfer function

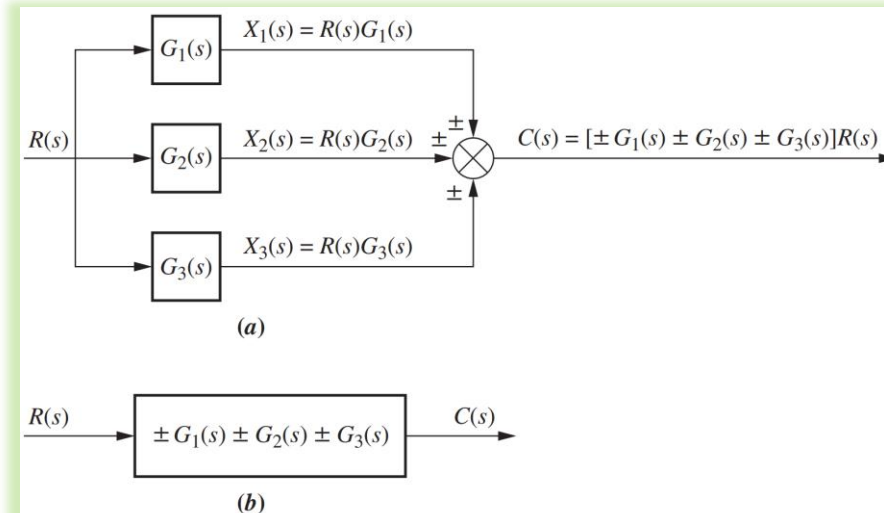


FIGURE 5.5 a. Parallel subsystems; b. equivalent transfer function

Block diagram reduction



Feedback

$$G_e(s) = \frac{G(s)}{1 \pm G(s)H(s)}$$

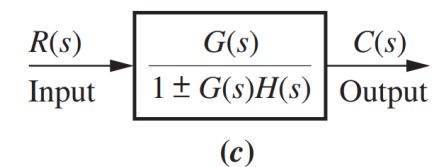
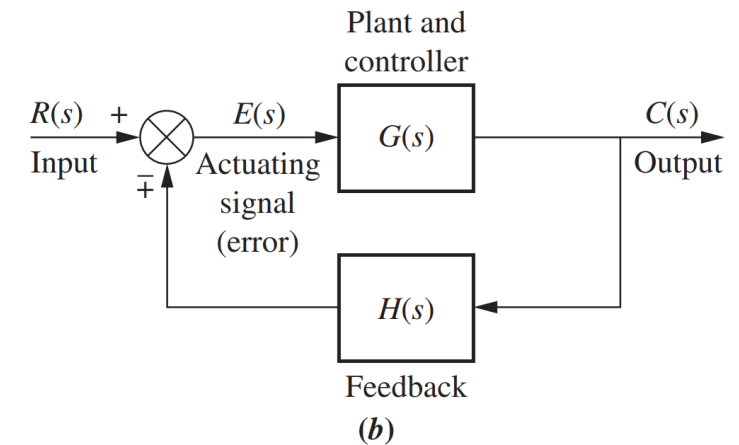
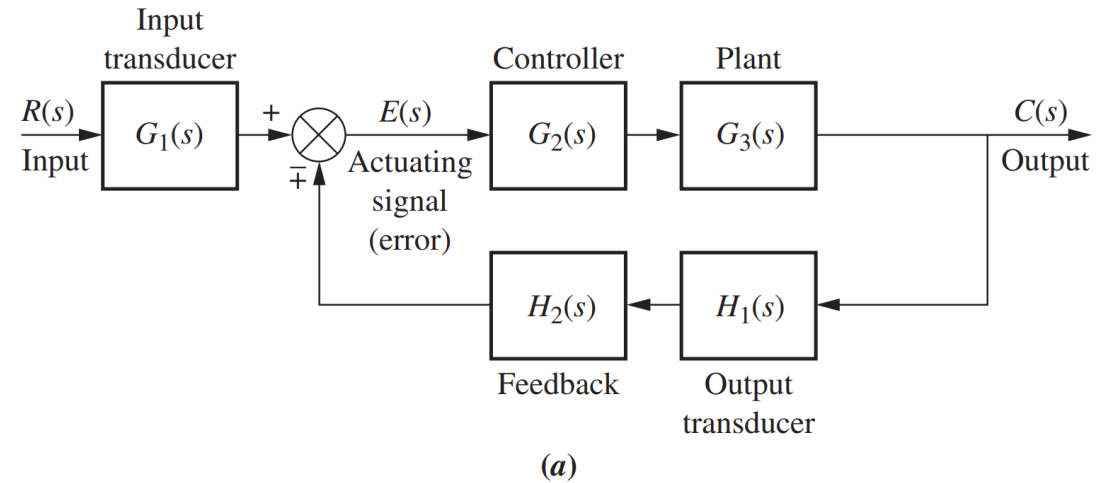
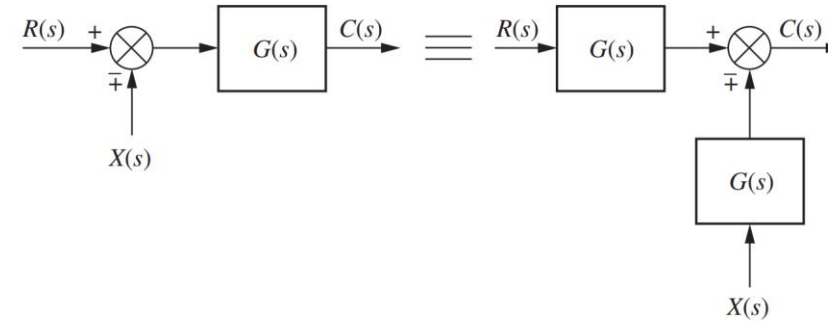
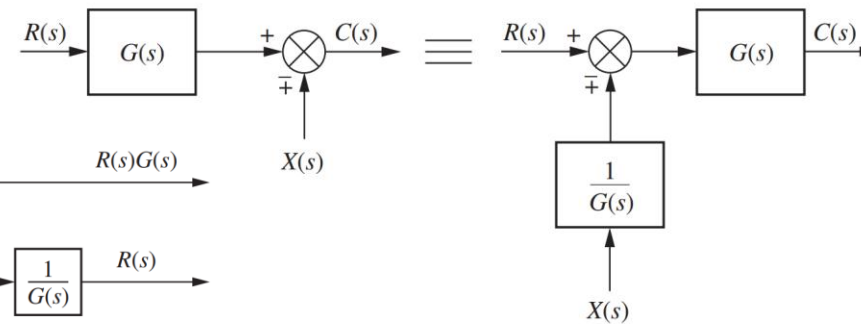


FIGURE 5.6 **a.** Feedback control system; **b.** simplified model; **c.** equivalent transfer function

□ Moving blocks



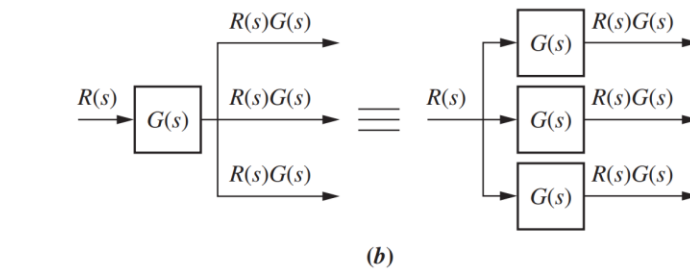
(a)



(b)

FIGURE 5.7 Block diagram algebra for summing junctions—equivalent forms for moving a block **a.** to the left past a summing junction; **b.** to the right past a summing junction

FIGURE 5.8 Block diagram algebra for pickoff points—equivalent forms for moving a block **a.** to the left past a pickoff point; **b.** to the right past a pickoff point



(b)

PROBLEM: Reduce the block diagram shown in Figure 5.9 to a single transfer function.

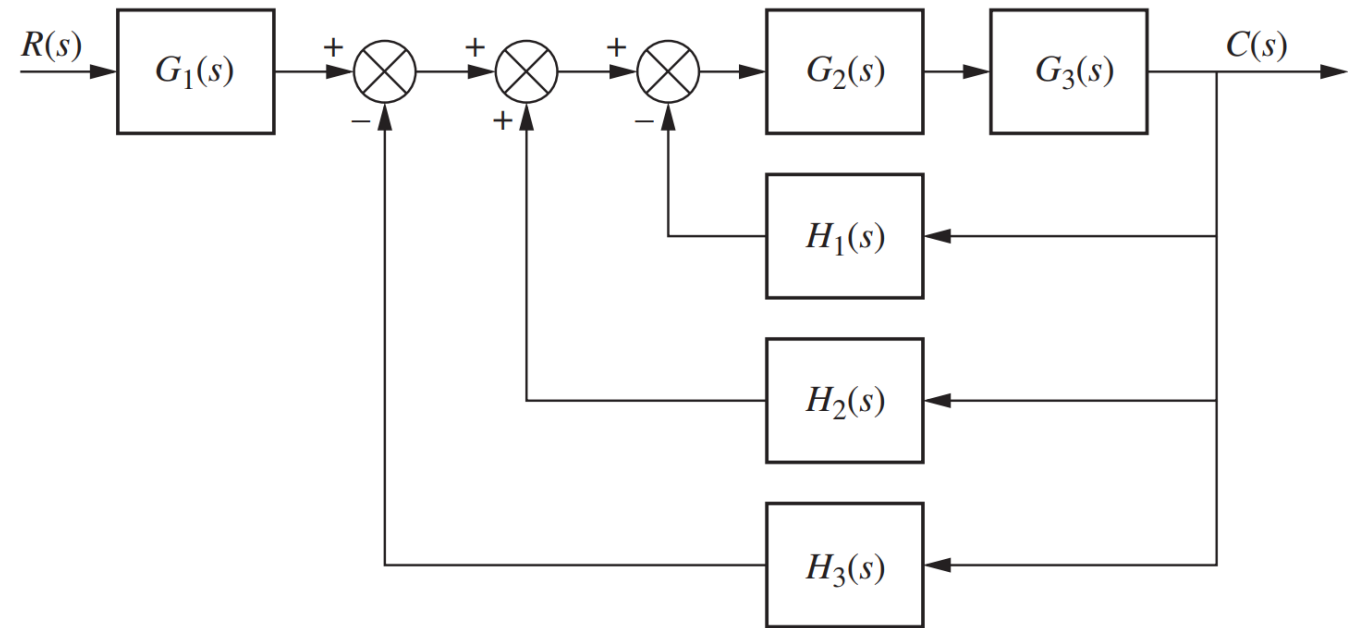


FIGURE 5.9 Block diagram for Example 5.1

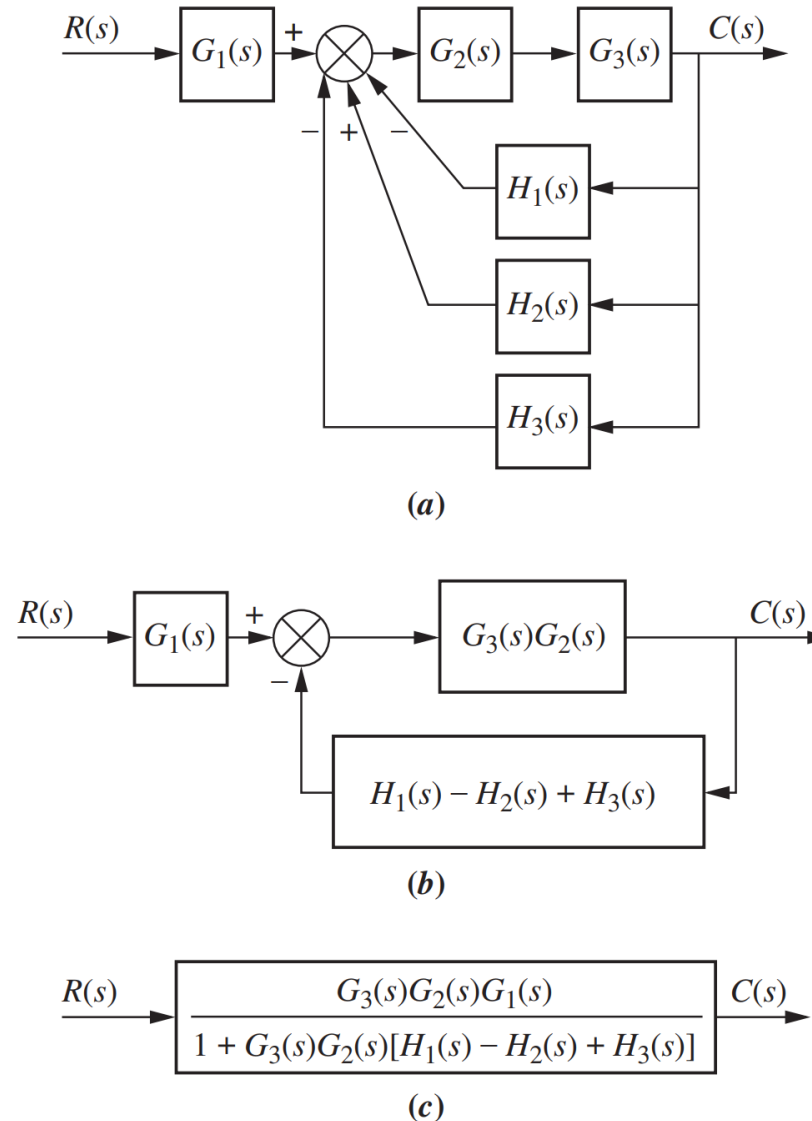


FIGURE 5.10 Steps in solving Example 5.1:
a. Collapse summing junctions;
b. form equivalent cascaded system in the forward path and equivalent parallel system in the feedback path; **c.** form equivalent feedback system and multiply by cascaded $G_1(s)$

Block Diagram Reduction by Moving Blocks

PROBLEM: Reduce the system shown in Figure 5.11 to a single transfer function.

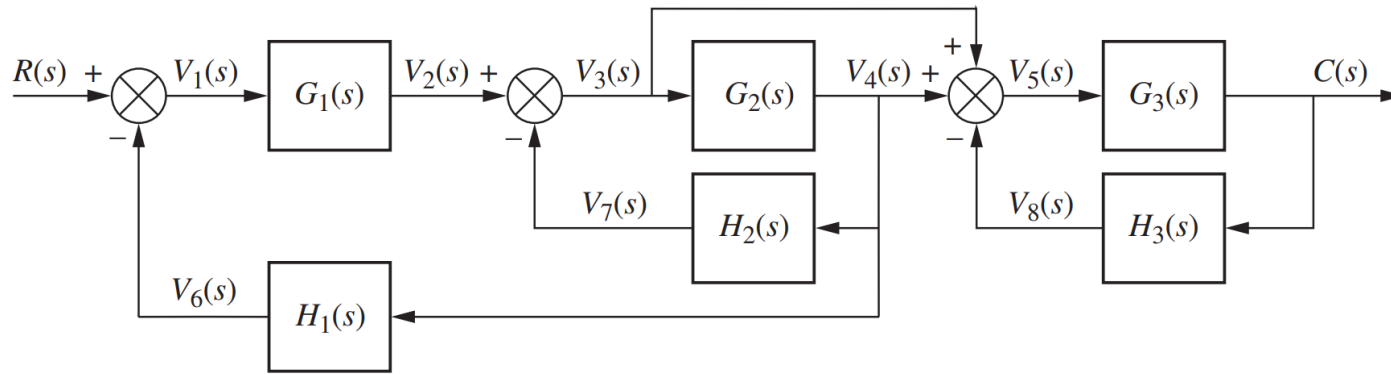
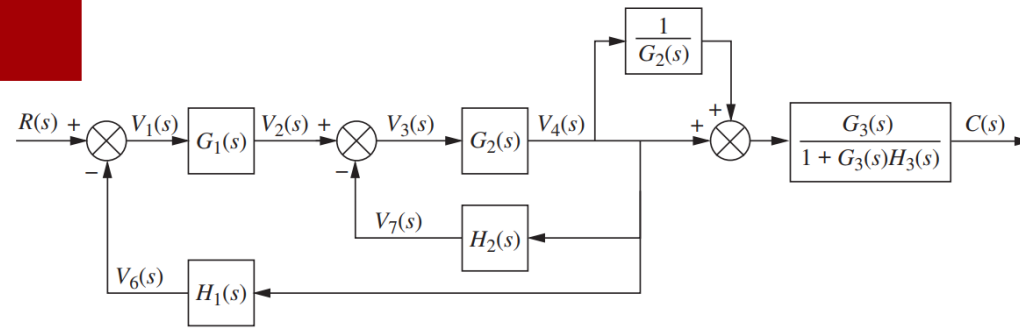
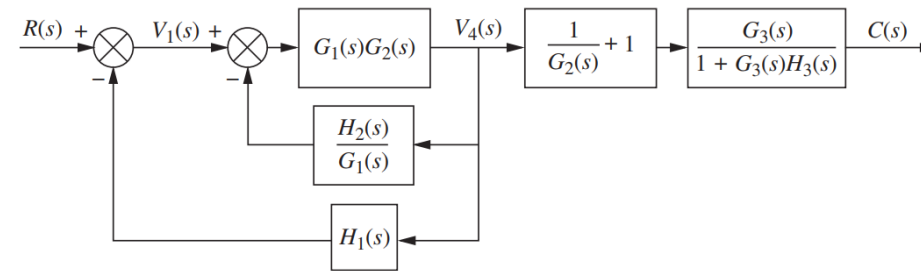


FIGURE 5.11 Block diagram for Example 5.2

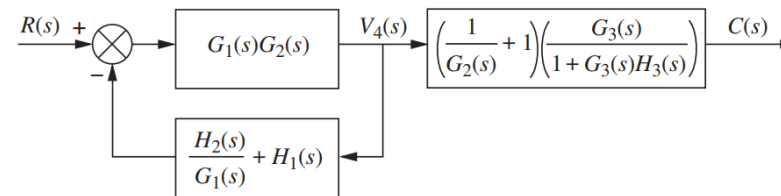
Block diagram reduction



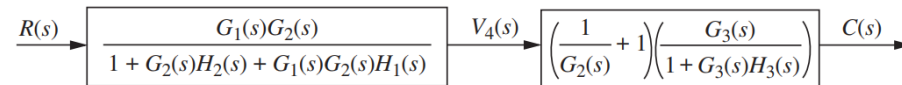
(a)



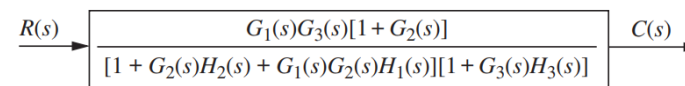
(b)



(c)



(d)



(e)

FIGURE 5.12 Steps in the block diagram reduction for Example 5.2

Block diagram reduction



PROBLEM: Find the equivalent transfer function, $T(s) = C(s)/R(s)$, for the system shown in Figure 5.13.

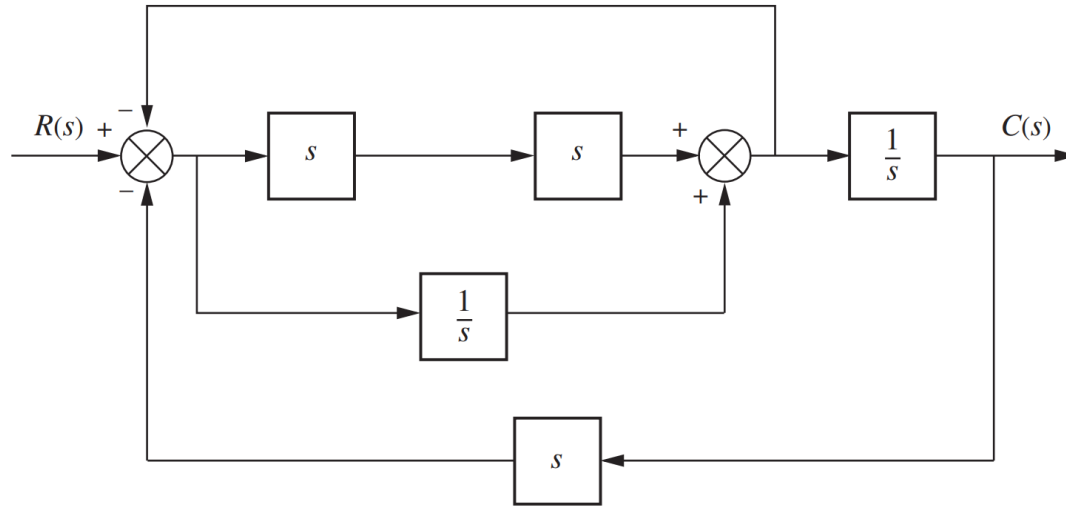


FIGURE 5.13 Block diagram for Skill-Assessment Exercise 5.1

ANSWER:



TryIt 5.1

Use the following MATLAB and Control System Toolbox statements to find the closed-loop transfer function of the system in Example 5.2 if all $G_i(s) = 1/(s + 1)$ and all $H_i(s) = 1/s$.

```
G1=tf(1,[1 1]);
G2=G1; G3=G1;
H1=tf(1,[1 0]);
H2=H1; H3=H1;
System=append(...
    (G1, G2, G3, H1, H2, H3);
input=1; output=3;
Q=[ 1  -4   0   0   0
    2   1  -5   0   0
    3   2   1  -5  -6
    4   2   0   0   0
    5   2   0   0   0
    6   3   0   0   0];
T=connect(System,...
    Q, input, output);
T=tf(T); T=minreal(T)
```

System Model

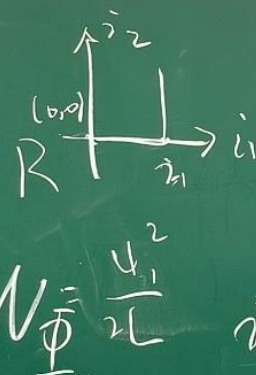
1. O.D.E $\rightarrow$ initial conditions set to zero $\left(\frac{Y(s)}{R(s)} = \frac{S^2}{s+1} \right)$ (not proper)
2. Transfer Function (only describe a pair of I/O relation)
3. Block Diagram.



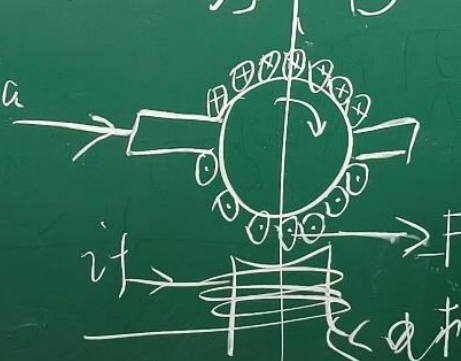
4. Signal Flow Graph

$$d\psi_1(i_1, i_2) = \frac{\partial \psi_1}{\partial i_1} di_1 + \frac{\partial \psi_1}{\partial i_2} di_2$$

$$\psi_1 = L_1 i_1 + L_2 i_2$$



Force-Voltage
Analogy



$$U - Ri = L \frac{di}{dt} \rightarrow \text{Faraday's Law}$$

$$T = J \frac{d\omega}{dt} = K_m i_f(t) - b\omega$$

$$T_m \dot{p} = m \dot{v}$$

$$e = -\frac{d\psi}{dt} = -\frac{d(Li)}{dt}$$

$$= \frac{dL}{dt} i + L \frac{di}{dt}$$

$$= \frac{\partial L}{\partial \phi} \omega i + L \frac{di}{dt}$$